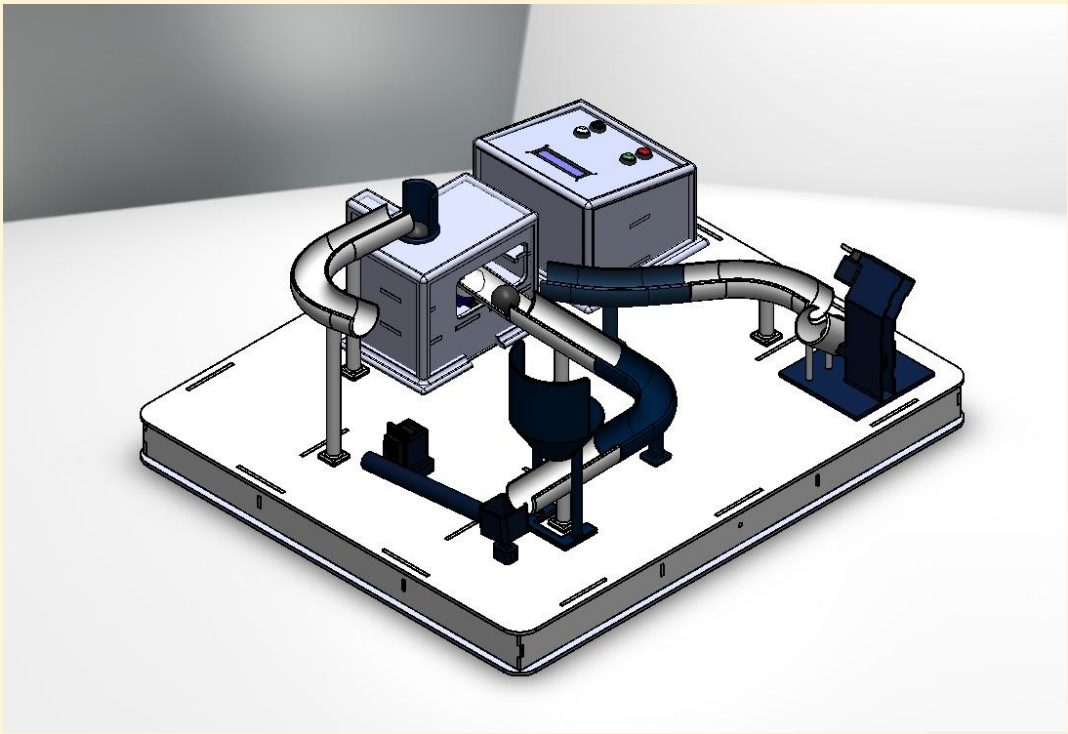
Ain Shams University — Faculty of Engineering

Mechatronics Engineering and Automation

AMONG US



Final Project — Technical Report -TEAM_9

MCT231 — Basics of Mechatronics

Academic Year 2025 / 2026

Supervisor: Associate Prof. Dr. Shady A. Maged

Team Members

This project was designed, built, and documented by the following team. Roles are indicated to reflect each member's primary contribution across the design, fabrication, electronics, firmware, and documentation workstreams.

Name	Primary Role	Student ID
Omar Mohamed Hegazy	Coding + CAD design	24P0121
Anas Mohamed Mohamed Galal	CAD design	24P0390
Abdellrahman Abdellateef	Application, communication, code integration	24P0161
Ahmed Essa Soliman	PCB, wiring diagrams, PDS, block diagram	24P0407
Mario Ashraf Sami	Calculations + documentation	24P0100

Project repository:

[GITHUB REPOSITORY LINK]



Abstract

AMONG US is an interactive, activity-based mechatronics toy built around a fully physical closed-loop marble pathway. The toy challenges a player to “defuse a sabotaged lunar reactor” by reproducing a colour sequence generated by a companion mobile application. White and black balls represent pure fuel cells; the player enters a candidate code on a physical control panel, and the machine then physically validates each ball in turn by sorting it on colour, routing matches to a projectile shooter and mismatches down a return path, and finally lifting every ball back to the start so the cycle can repeat without manual reloading.

The system is implemented with a two-microcontroller master/slave architecture. An Arduino Mega serves as the master, hosting the user interface (an I2C 16×2 LCD and physical buttons), the game logic, and the HC-05 Bluetooth link to the Processing application. An Arduino Uno serves as the slave, driving three servos, an L298N-controlled DC shooter motor, and the infrared presence and colour sensors. The two boards communicate over a UART serial link.

This report documents the full design-thinking process — from user research and ideation through mechanical and electrical design, engineering calculations, prototyping, testing, and final integration — and maps each deliverable against the MCT231 assessment rubric.

Table of Contents

Team Members 2
Abstract 3
Table of Contents 4
List of Figures & Tables ..... **Error! Bookmark not defined.**
1. Introduction 7
1.1 Concept 7
1.2 Problem Statement 7
2. Project Theme & Story 8
2.1 Narrative 8
2.2 Visual Identity 8
3. User Research 9
3.1 Target User 9
3.2 Methods 9
3.3 Findings ..... **Error! Bookmark not defined.**
3.4 User Persona ..... **Error! Bookmark not defined.**
4. Design Challenge 10
4.1 Derived Requirements 10
5. Ideation & Concept Development 11
5.1 Mind Map ..... **Error! Bookmark not defined.**
5.2 Mood Board ..... **Error! Bookmark not defined.**
5.3 Concept Sketches ..... **Error! Bookmark not defined.**
5.4 Concept Selection 11
6. System Overview 12
6.1 Block Diagram 12
6.2 Closed-Loop Mechanical Flow 13
6.3 System Flowchart 14
7. Mechanical Design & Functionality 15
7.1 Sorting Mechanism 16
7.2 Shooting Mechanism 17
7.3 Lifting Mechanism 18
7.4 Chassis, Tracks & Materials 18
8. Electrical & Control Design 19
8.1 Control Architecture 19
8.2 Sensors 19
8.3 Actuators & Drivers 19

8.4 Power Distribution & PCB 20
8.5 Start / Stop (Emergency) Button 21
8.6 Wiring Diagrams 21
9. Firmware & Software 23
9.1 Master Firmware (Arduino Mega) 23
9.2 Slave Firmware (Arduino Uno) 23
9.3 Communication Protocol 24
9.4 Processing Application (Second Interface) 25
10. Engineering Calculations 26
10.1 Calculation Methodology (Formula Chain) 26
10.1.1 Projectile motion (target → launch speed) 26
10.1.2 Arm kinematics 26
10.1.3 Trapezoidal profile 26
10.1.4 Inertia 26
10.1.5 Phase torques 26
10.1.6 RMS torque & safety factor 26
10.1.7 L298N DC-motor electrical sizing (shooter) 26
10.1.8 Servo adequacy check (lift & sorting) 27
10.2 Shooting Mechanism — Projectile 27
10.3 Actuator Sizing — Shooting (DC Motor) 27
10.4 Actuator Sizing — Sorting Servos 28
10.4.1 Upper Sorting Servo (Servo 1) 28
10.4.2 Lower Sorting Servo (Servo 2) 28
10.5 Actuator Sizing — Lifting Servo 29
10.6 Current Consumption & Power Sizing 29
11. Prototyping & Iteration 33
11.1 Iteration Log 33
12. Safety Considerations 34
13. Educational Impact 35
14. Project Management 36
14.1 Team Roles & Responsibilities 36
14.2 Work Breakdown Structure (WBS) 37
14.3 Gantt Chart (12-week) 38
14.4 Product Design Specification (PDS) 39
15. Bill of Materials (BOM) 40
16. Assurance of Properties 41

17. Final Product.....42
18. Conclusion & Future Work ..... **Error! Bookmark not defined.**
18.1 Known Open Items ..... **Error! Bookmark not defined.**
18.2 Future Improvements..... **Error! Bookmark not defined.**
References ..... **Error! Bookmark not defined.**
Appendix A — Engineering Drawings & CAD..... **Error! Bookmark not defined.**
Appendix B — Additional Documentation..... **Error! Bookmark not defined.**

1. Introduction

AMONG US is an electromechanical toy for children aged 2–10, built for the MCT231 final project. It integrates three mechanisms — colour sorting, projectile shooting, and vertical lifting — into a single closed-loop marble path with one entry point. Control is split across two communicating microcontrollers (Arduino Mega and Arduino Uno), and two interfaces report state: an on-board I2C 16×2 LCD and a Bluetooth-linked Processing application.

1.1 Concept

The toy is framed as a reactor-defusal game inspired by the cooperative game Among Us. The player filters pure fuel cells (white and black balls) from corrupted ones to stabilise a sabotaged lunar reactor before a countdown expires. Mechanically the task is colour matching and material handling; the theme supplies the objective and time pressure.

1.2 Problem Statement

- Many electronic toys for this age group are either screen-only or mechanical with no feedback loop.
- Objective: a safe, engaging toy that physically demonstrates an automated sort–transport–dispatch process.
- Skills targeted: colour matching, sequencing, pattern recognition, and cause-and-effect reasoning.

2. Project Theme & Story

2.1 Narrative

A lunar-base reactor has been sabotaged and must be stabilised by feeding pure fuel cells in the correct order. The companion app generates a target colour sequence, runs a 120-second countdown, and tracks lives. A correct cell advances the defusal; an incorrect cell costs a life and adds a time penalty.

2.2 Visual Identity

The palette is dark-space: navy blue and lunar grey as primaries, cyan as the accent, and red reserved for the countdown and error states.

Element	Specification
Primary colours	Deep navy blue, lunar grey
Accent / lighting	Cyan neon
Chassis material & finish	Laser-cut MDF
Track material & finish	3D-printed (FDM), navy blue

3. User Research

3.1 Target User

- Children aged 2–10; the code-entry interaction targets the upper band (~6–10).
- Motion, lights, and the shooter keep the toy engaging for younger children under supervision.

3.2 Methods

- Parent interviews.
- Observation of children using comparable toys.
- Review of comparable products and children’s-toy safety standards.

4. Design Challenge

Design a safe, closed-loop toy that demonstrates an automated sort–transport–dispatch process while reinforcing colour matching and sequencing for the target age group.

4.1 Derived Requirements

Combined with the course brief, the challenge yields the following requirements (tracked in the PDS, Section 14.3):

- Three functional mechanisms: sorting, shooting, lifting.
- Closed loop, single marble entry point, no manual reloading during play.
- ≥ 2 microcontrollers communicating over a defined protocol.
- Two interfaces presenting two different sets of information.
- ≥ 1 custom PCB, single power source, hidden wiring, physical start/stop button.
- Children’s-toy safety compliance; footprint within the course size limit.

5. Ideation & Concept Development

Ideation explored themes, mechanisms, and the play loop before converging on the reactor-defusal concept. Supporting artefacts follow.

5.1 Concept Selection

Alternatives for each subsystem (sorting method, shooter type, lift design) were compared in a weighted decision matrix, addressing the “evaluate alternative design solutions” outcome.

6. System Overview

6.1 Block Diagram

A single power source feeds a DC buck converter that distributes regulated rails through the custom PCB to both microcontrollers and all peripherals. The Mega handles the user interface and communication; the Uno handles all sensing and actuation. The two boards exchange short serial messages.

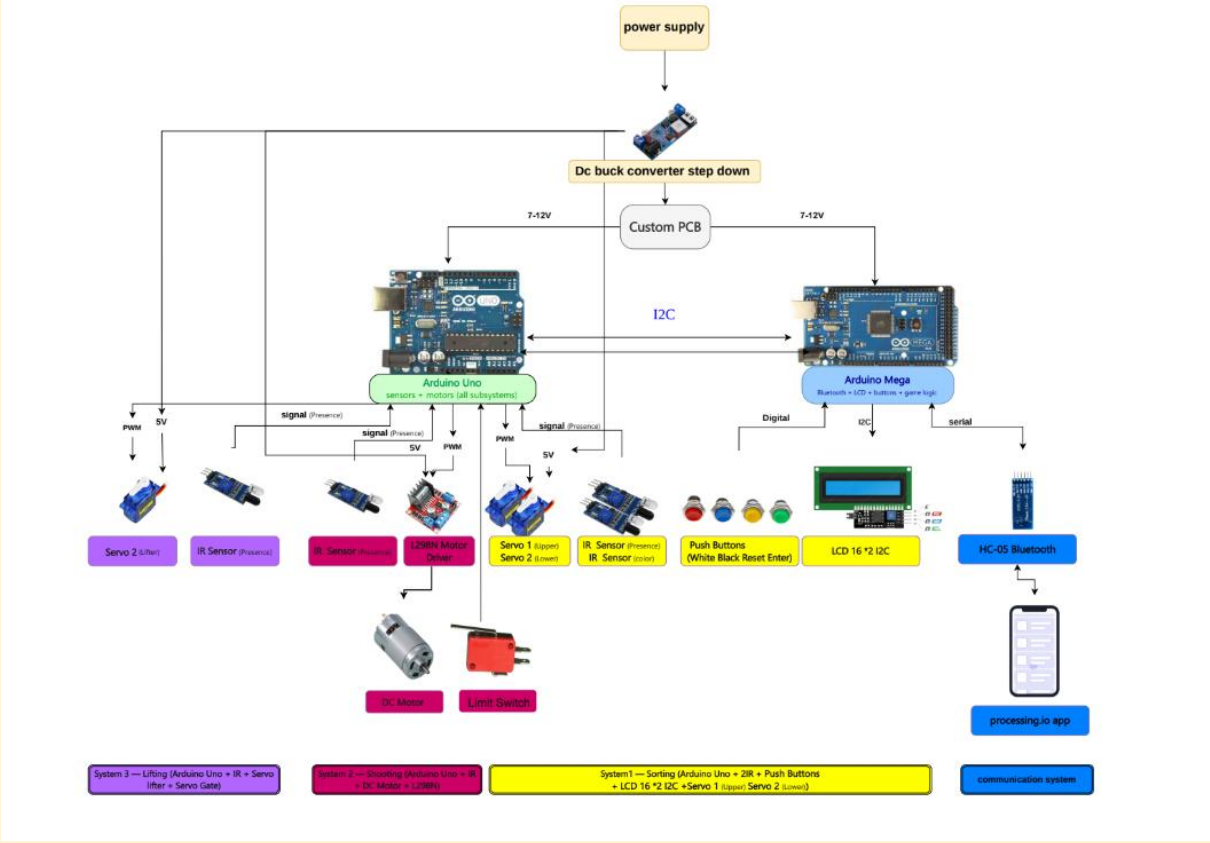


Figure 6.1 — Whole-system block diagram.

6.2 Closed-Loop Mechanical Flow

Each ball follows one continuous path and never leaves the machine during play:

1. Sorting: an IR colour sensor reads the ball at the entry point.
2. Match: the path servo routes the ball to the shooter, which fires it into the collection funnel; the funnel feeds it to the lift.
3. Mismatch: the path servo routes the ball down the return path, bypassing the shooter, directly to the lift.
4. Lift: the lifting servo raises the ball back to the sorting stage, closing the loop.

The app maintains the target sequence, the 120-second countdown, and the life counter, so the physical loop and game state advance together.

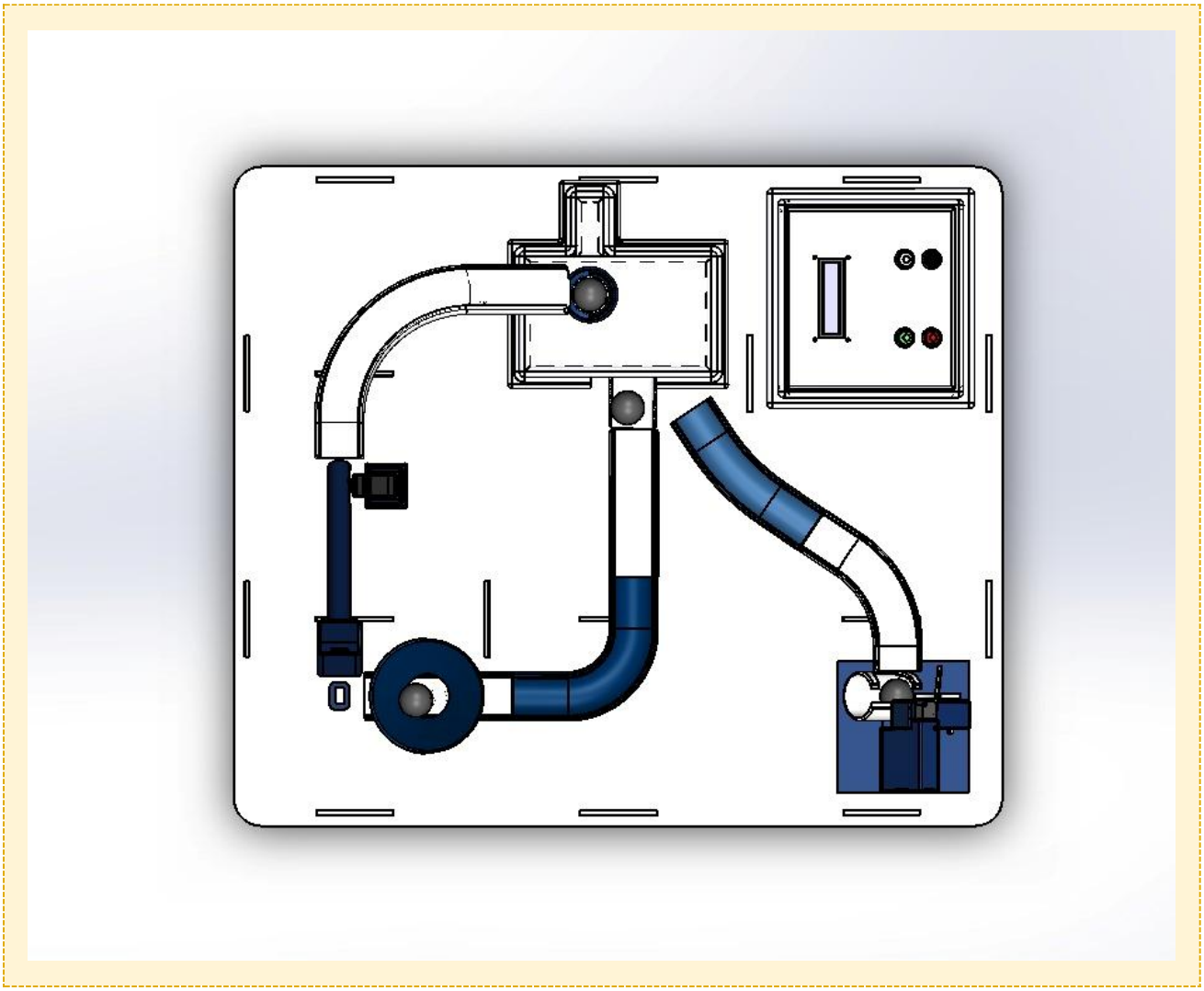


Figure 6.2 — Closed-loop marble path.

6.3 System Flowchart

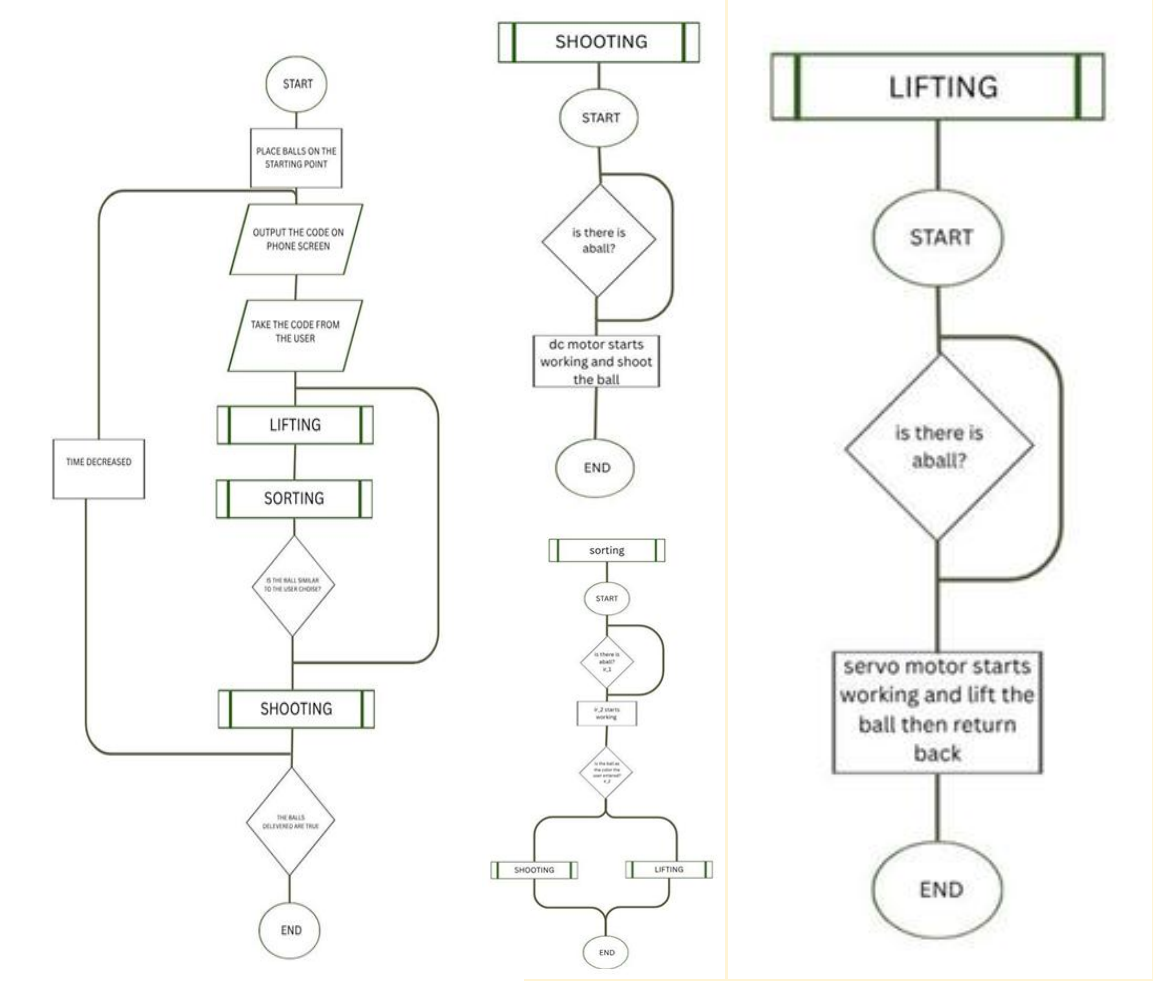


Figure 6.3 — System operation flowchart.

7. Mechanical Design & Functionality

The mechanical system comprises three subsystems connected by 3D-printed tracks into a single closed loop. All three are mounted on a laser-cut MDF baseboard. This section describes each mechanism, its actuators, and its role in the loop. CAD models were produced in SolidWorks.

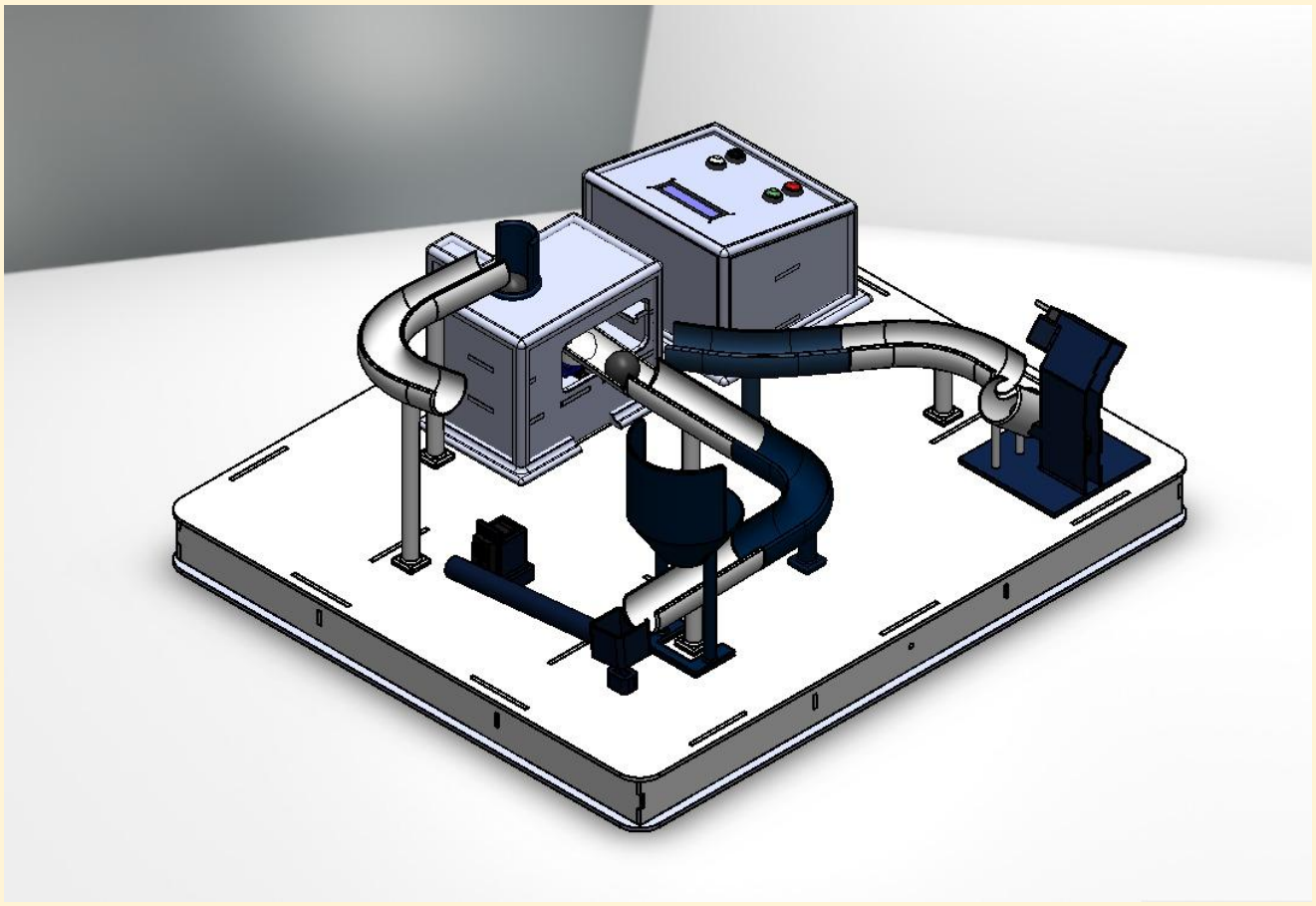


Figure 7.1 — Full integrated CAD assembly.

7.1 Sorting Mechanism

The sorting mechanism is the entry point of the loop. A ball arriving from the lift drops into a holder. A presence IR sensor detects the ball; the colour IR sensor is then powered briefly to read the ball as white or black. Based on whether the reading matches the current slot in the target sequence, a path servo routes the ball either toward the shooter or down the return path, and a holder servo releases the ball into the selected path.

Two servos are used in this subsystem — a holder/release servo and a path-selection servo — driven by the Uno. The colour sensor's emitter is switched on only during a reading to improve repeatability and reduce stray light effects.

Servo	Function	Key positions (degrees)
Holder servo	Holds then releases the ball	Home 180° → Reach 90°
Path servo	Selects shooter vs return path	Home 105° / Shooter 142° / Return 97°

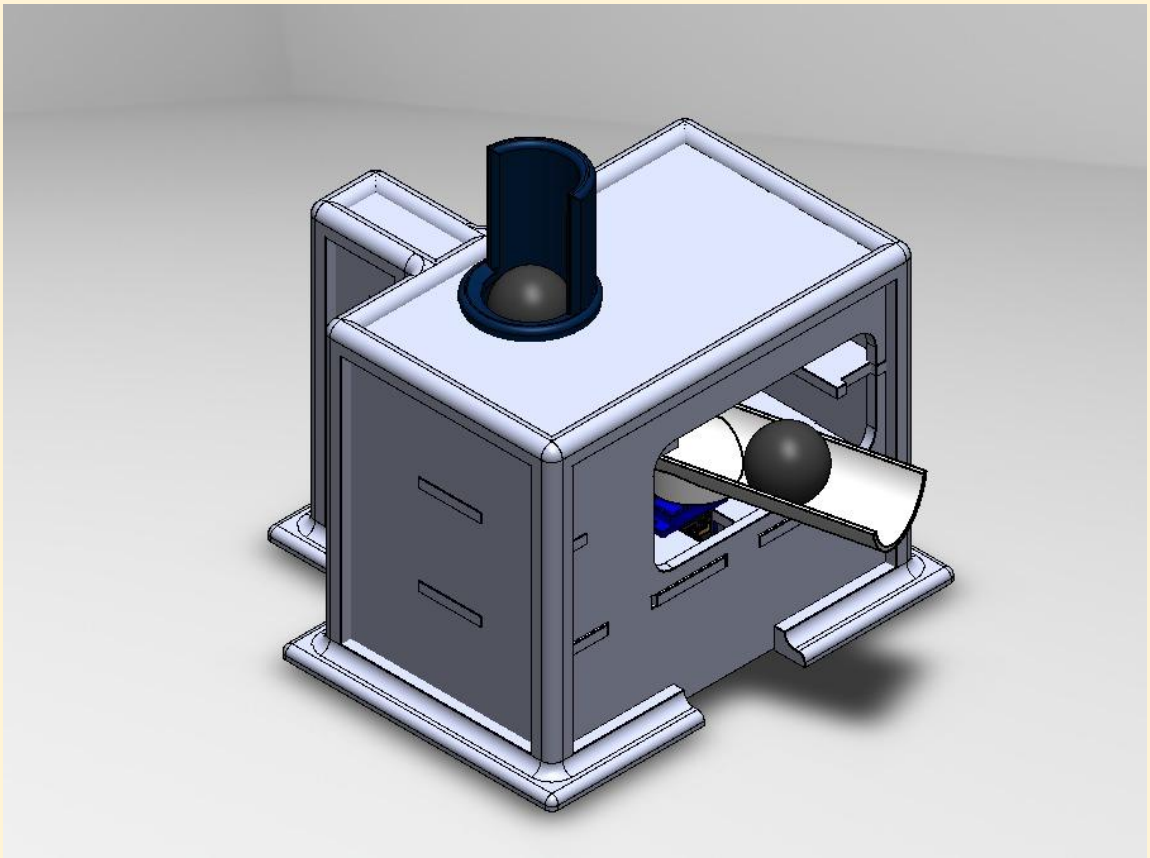


Figure 7.2 — Sorting mechanism.

7.2 Shooting Mechanism

When a ball is accepted, it rolls to the shooter barrel, where a shooter IR sensor confirms arrival. A DC motor driven through an L298N module then fires the ball forward into the collection funnel. The motor runs at full speed to launch the ball, then reverses at a slow speed to retract until a limit switch is reached, which re-homes the firing arm and prepares it for the next shot.

The projectile design target is a horizontal range of approximately 30 cm at a 45° launch angle; the supporting calculations are in Section 10.1.

Parameter	Value
Actuator	DC motor via L298N driver (PWM speed control)
Firing speed (PWM)	94 / 255
Retract speed (PWM)	35 / 255
Homing	Limit switch at retracted position
Arrival detection	IR sensor at barrel

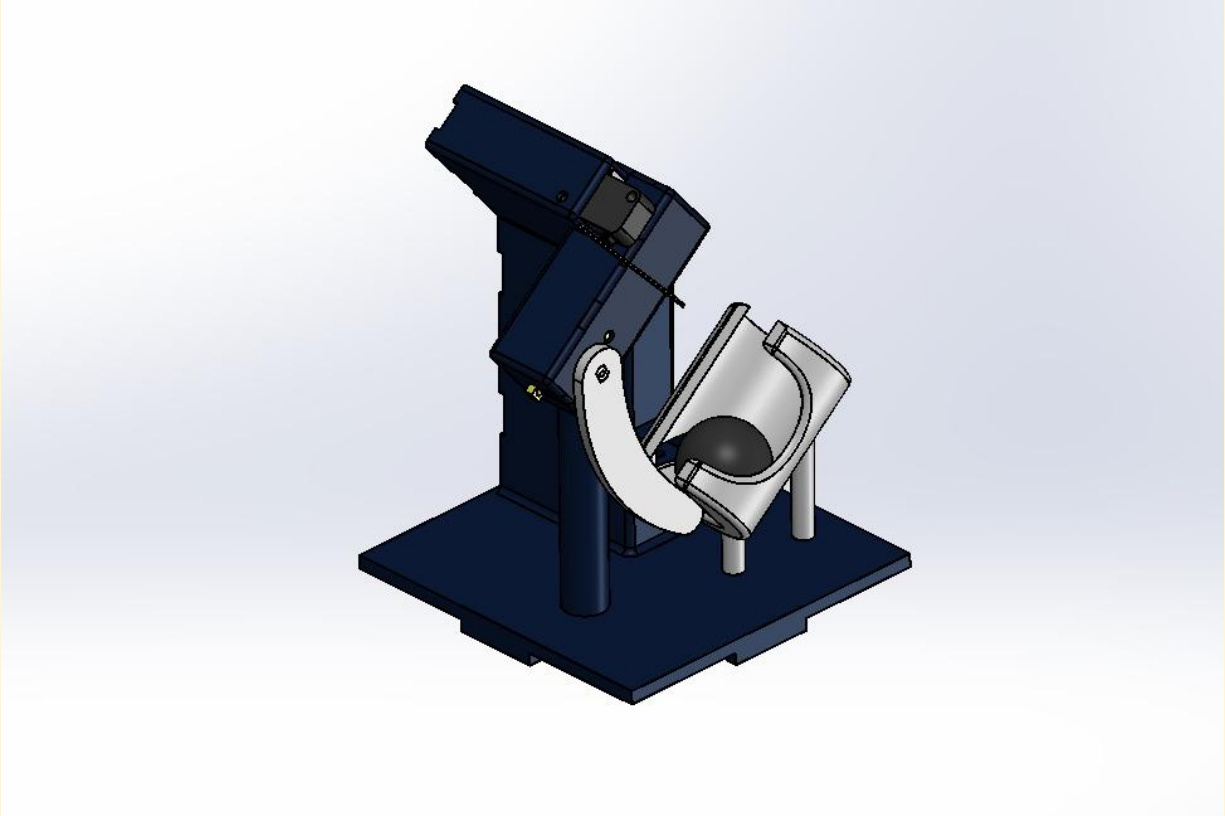


Figure 7.3 — Shooting mechanism.

7.3 Lifting Mechanism

The lifting mechanism is the common return stage for both branches of the loop. A lift IR sensor detects a ball arriving at the elevator; after a short settling delay, the lift servo raises the ball and then returns to its rest position, delivering the ball back to the sorting mechanism. Because both the shot balls (via the funnel) and the rejected balls (via the return path) arrive here, the lift guarantees that every ball re-enters the loop.

Parameter	Value
Actuator	Lift servo
Rest position	0°
Lift position	95°
Arrival detection	IR sensor at elevator base

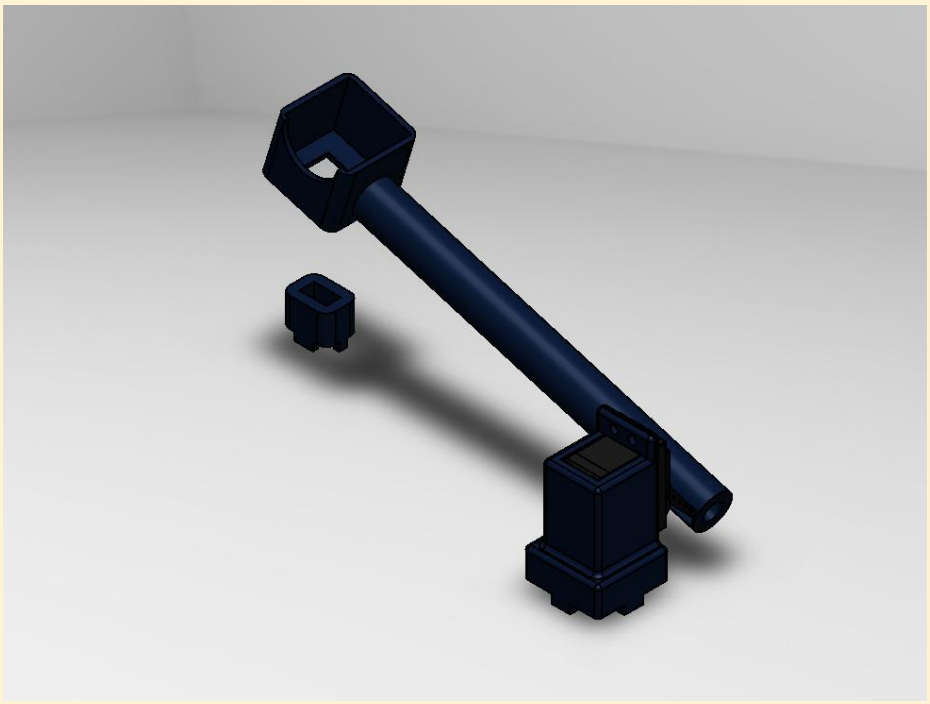


Figure 7.4 — Lifting mechanism.

7.4 Chassis, Tracks & Materials

The baseboard and structural panels are laser-cut MDF, painted matte lead grey. The marble tracks and funnels are FDM 3D-printed in navy-blue filament. The design is intended to be modular: each subsystem has its own mounting and wiring so it can be removed and tested independently (see the maintenance requirement in the PDS, Section 14.3).

8. Electrical & Control Design

8.1 Control Architecture

The system uses a master/slave architecture. The Arduino Mega is the master: it reads the control-panel buttons, drives the I2C LCD, hosts the game logic and final verification, and manages the HC-05 Bluetooth link to the app. The Arduino Uno is the slave: it reads the IR sensors and drives the three servos and the DC shooter motor. The boards are linked over UART (see Section 9.3 for the protocol).

Board	Responsibilities	Key peripherals
Arduino Mega (master)	UI, game logic, comms, verification	I2C LCD, buttons, HC-05, BT state pin
Arduino Uno (slave)	Sensing & actuation	2× IR (presence/colour), holder/path/lift servos, L298N + DC motor, limit switch, barrel IR

8.2 Sensors

Sensor	Purpose	Interface
IR presence (sorting)	Detect ball in holder	Digital, INPUT_PULLUP
IR colour (sorting)	Read white vs black	Digital, emitter power-gated
IR barrel (shooter)	Confirm ball at barrel / left chamber	Digital, INPUT_PULLUP
Limit switch (shooter)	Home the firing arm	Digital, INPUT_PULLUP
IR lift	Detect ball at elevator	Digital, INPUT_PULLUP
HC-05 state pin (Mega)	Detect Bluetooth disconnection	Digital

Note: the colour reading uses a single IR sensor with its emitter switched on only during measurement. If reliability of white/black discrimination proves marginal in testing, document any calibration or threshold tuning here.

8.3 Actuators & Drivers

Actuator	Driver	Control
Holder servo	Direct from Uno	PWM angle
Path servo	Direct from Uno	PWM angle
Lift servo	Direct from Uno	PWM angle
DC shooter motor	L298N H-bridge	PWM (enA) + direction (in1/in2)

8.4 Power Distribution & PCB

The system is powered from a single SMPS (220 V AC → 12 V DC). The 12 V rail is distributed by the custom PCBs and split three ways: directly to the L298N for the DC shooter motor, through a buck converter to the servo rail, and through a 9 V linear regulator to the Uno's VIN. All rails share a common ground. The design uses two custom PCBs implemented as shields — one for the Uno and one for the Mega — with the Arduino boards seated on pin headers. No breadboards are used in the final build.

Item	Specification
Source	Single SMPS, 220 V AC → 12 V DC
12 V rail	→ L298N → DC shooter motor
Servo rail	12 V → buck converter → servos
Logic rail	12 V → 9 V linear regulator → Uno VIN
PCBs	Two custom shield-PCBs (Uno + Mega) on pin headers, common GND

Note: two custom PCBs exceed the "≥ 1 custom PCB" requirement. Confirm the Mega's supply path (its own regulator or VIN off the same 12 V rail) when finalising the schematic.

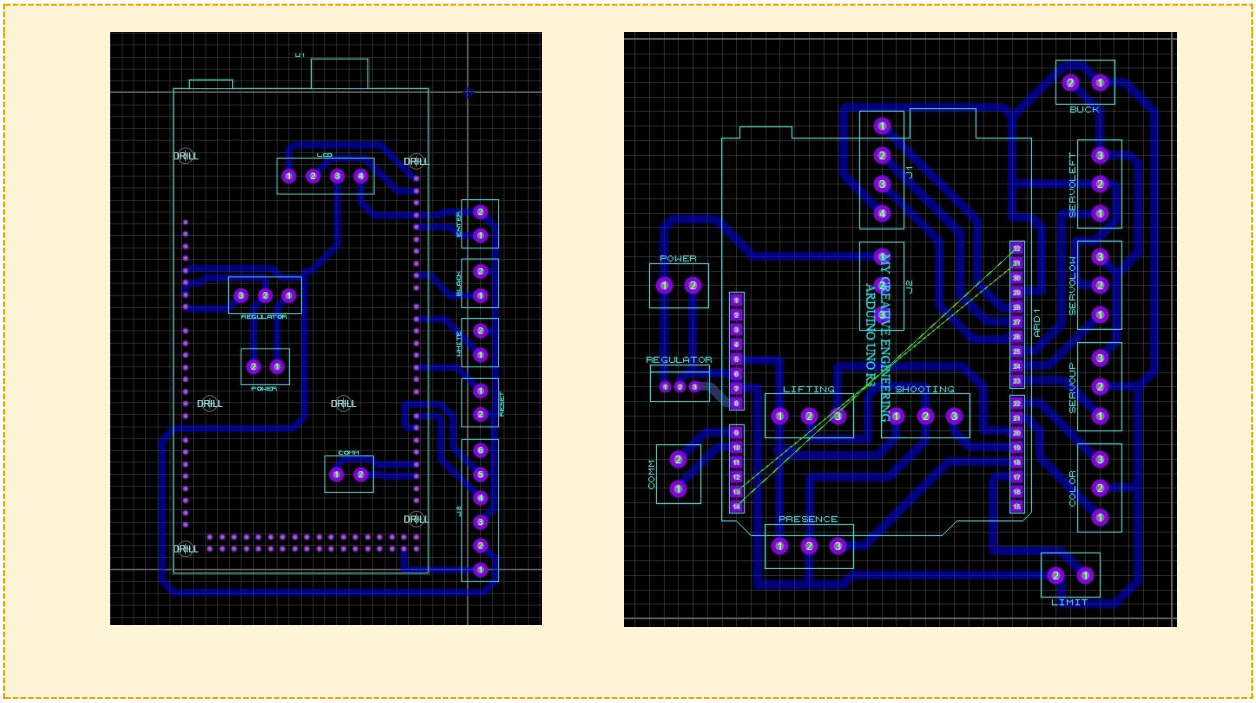


Figure 8.1 — Custom shield-PCBs.

Figure 8.2 — Manufactured PCB.

8.5 Start / Stop (Emergency) Button

A physical start/stop button provides a hardware-level means to halt the system. Describe its wiring (e.g. interrupting a rail or driving a reset path) and how the firmware responds to it. The firmware also supports a software reset command (“R”) that stops the motor, returns all servos to home, and clears the active sequence.

8.6 Wiring Diagrams

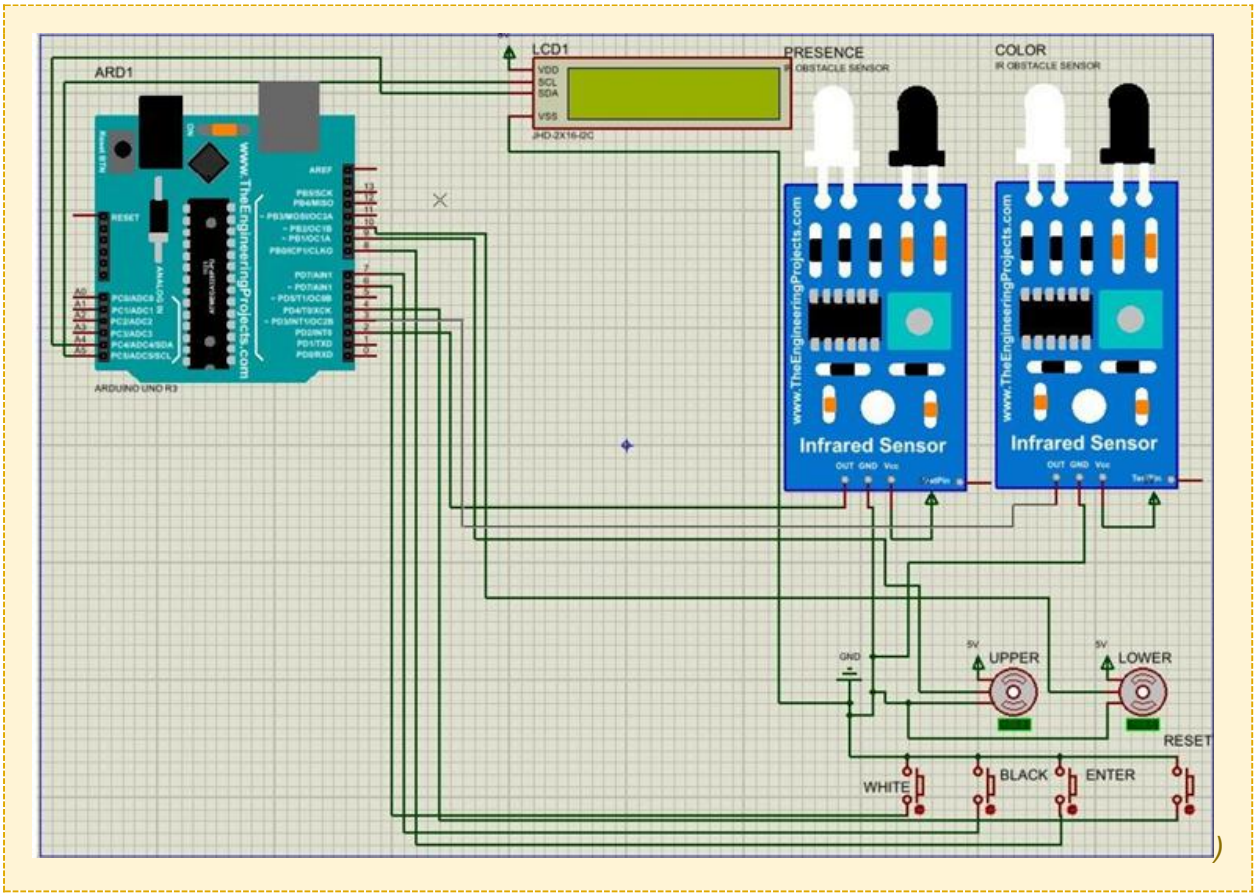


Figure 8.3 — Sorting wiring diagram.

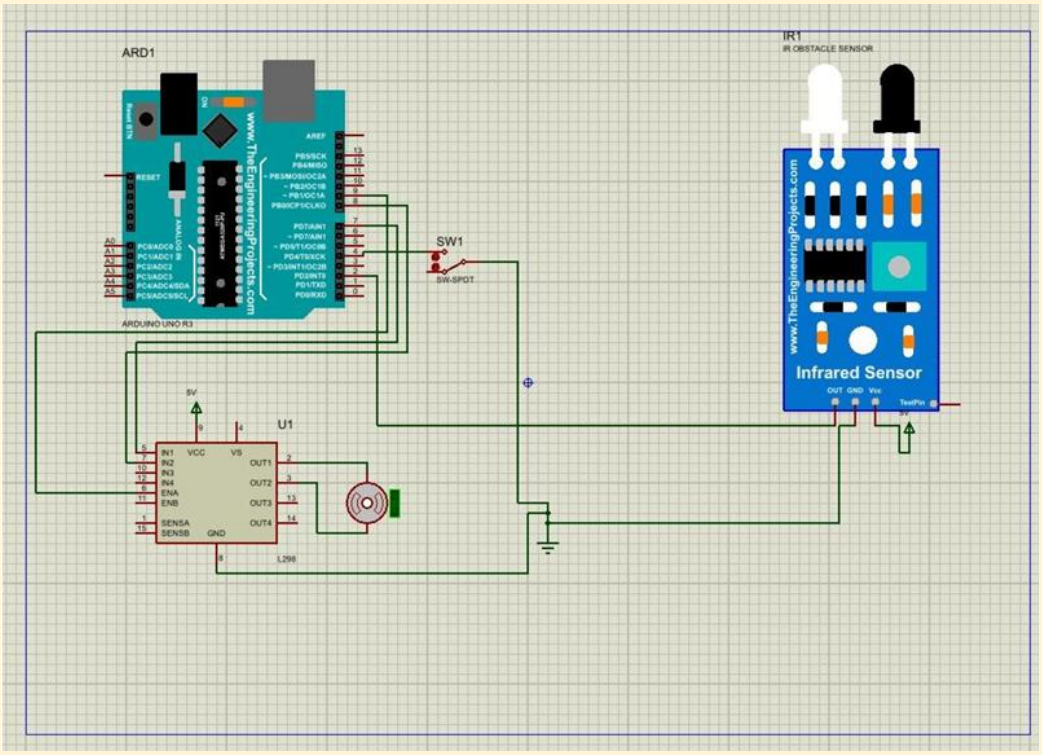


Figure 8.4 — Shooting wiring diagram.

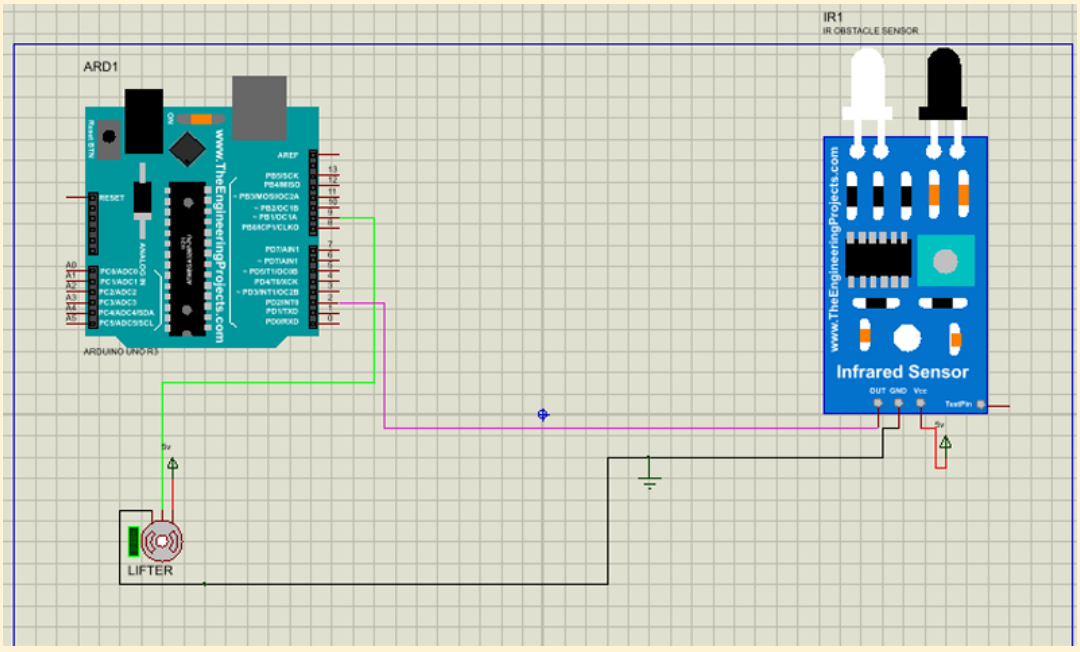


Figure 8.5 — Lifting wiring diagram.

9. Firmware & Software

The control software spans three programs: the Mega master firmware, the Uno slave firmware, and the Processing application. Full source listings are provided in the appendices; this section explains the design and the message protocol that ties them together.

9.1 Master Firmware (Arduino Mega)

The Mega runs a state machine with three states: awaiting the target code from the app, awaiting the player's button input, and awaiting the Uno to finish physical execution. It receives the app-generated target sequence over Bluetooth, accepts the player's code via the white/black/reset/submit buttons (mirroring it on the LCD), transmits the submitted code to the Uno, and on completion verifies the player's code against the target and reports success or failure to both the LCD and the app. It also includes two safety paths: a Bluetooth-disconnection failsafe (via the HC-05 state pin) and a remote reset from the app.

State	Behaviour
AWAIT_APP	Receive target colour sequence from the app over Bluetooth
AWAIT_INPUT	Read player buttons, display code on LCD, submit when complete
AWAIT_UNO	Wait for the Uno to report completion, then verify and report result

9.2 Slave Firmware (Arduino Uno)

The Uno receives the player's sequence from the Mega and then physically processes it ball by ball. For each ball it reads presence and colour, compares the colour against the current sequence slot, and either routes to the shooter (on a match — fire, then reload to the limit switch) or down the return path (on a mismatch). In both cases the ball is then lifted back to the start. A match advances to the next slot; a mismatch requires the same slot to be retried. When the whole sequence has been processed, the Uno signals completion to the Mega.

The Uno also honours the emergency reset command at multiple wait points (while waiting for the ball at the barrel and at the lift), so the system can be stopped mid-cycle and returned to a safe home state.

9.3 Communication Protocol

The Mega and Uno communicate over a UART serial link. Messages are short and single-purpose:

Direction	Message	Meaning
Mega → Uno	“WBW…” + newline	Player’s submitted colour sequence
Mega → Uno	“R”	Emergency / remote reset
Uno → Mega	“D”	All physical tasks complete
App → Mega	“W” / “B” chars	Target sequence generation
App → Mega	“R”	Remote reset
Mega → App	“C” / “W”	Verification result (correct / wrong)

Documentation note: the system block diagram currently labels the Mega–Uno link as I2C, while the firmware implements it as UART/serial. These should be reconciled so the report, block diagram, and PDS all describe the same protocol.

9.4 Processing Application (Second Interface)

The companion application is built with Processing. It generates the target colour sequence, runs the 120-second countdown, tracks lives, and applies time penalties on incorrect attempts. It communicates with the Mega over the HC-05 Bluetooth link.

Current status: the present version of the Processing sketch was developed and tested in a simulation environment, so the hardware Bluetooth pairing and serial-communication logic still needs to be integrated. Document the final app screens and the Bluetooth integration here once complete.

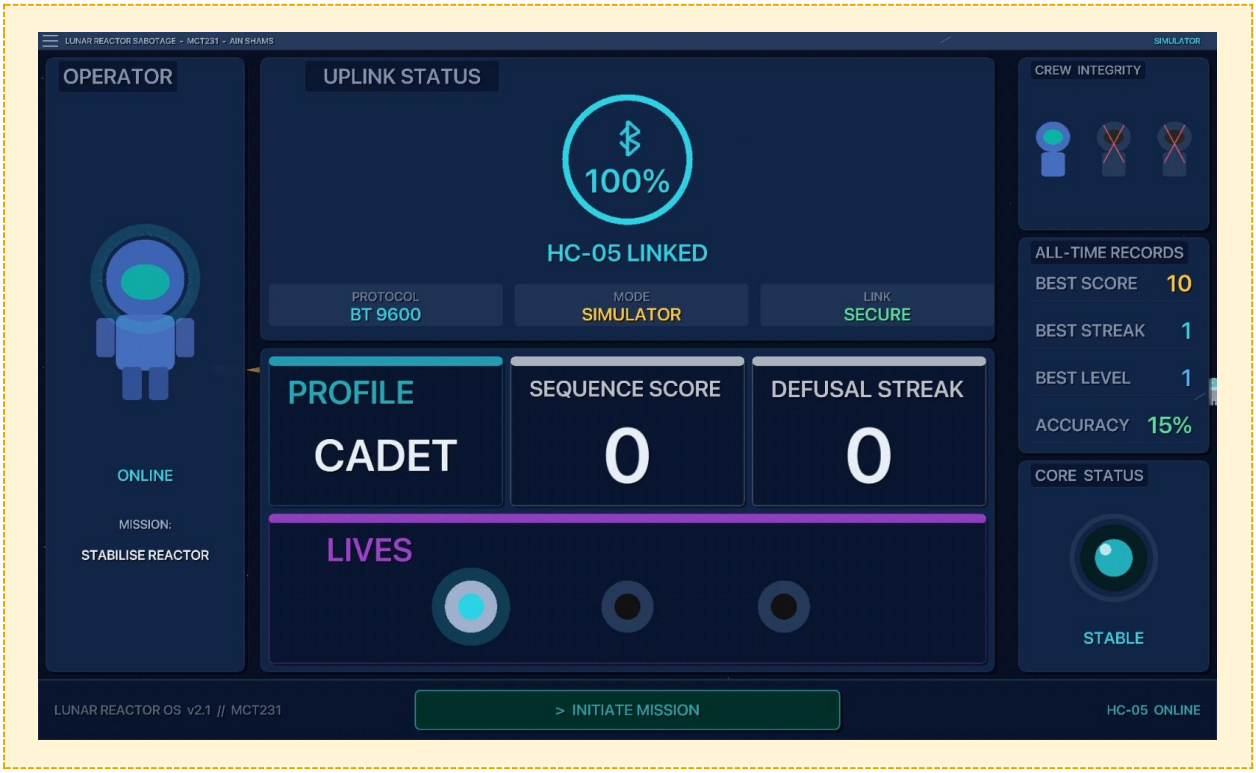


Figure 9.1 — Processing application interface.

10. Engineering Calculations

This section presents the projectile and actuator-sizing calculations for the system. Values are drawn from the team's CAD measurements and motion-profile assumptions; a safety factor of 1.5 is applied to the actuator torque results.

10.1 Calculation Methodology (Formula Chain)

All actuators are sized with the same chain: trapezoidal velocity profile → inertia → phase torques → RMS torque → safety factor. The shooter adds projectile motion and L298N DC-motor electrical sizing; the servos close with a stall-torque margin check.

10.1.1 Projectile motion (target → launch speed)

- Vertical rise: $\Delta H = y_{peak} - y_0$
- Vertical launch component: $v_y = \sqrt{(2 \cdot g \cdot \Delta H)}$
- Time of flight: $t_f = (v_y + \sqrt{(v_y^2 + 2 \cdot g \cdot y_0)}) / g$
- Horizontal component: $v_x = x / t_f$
- Launch speed: $v = \sqrt{(v_x^2 + v_y^2)}$

10.1.2 Arm kinematics

- $\omega_{max} = v / L$ (L = arm length); or for fixed motor speed N: $\omega_{max} = 2\pi N/60 \rightarrow L = v/\omega_{max}$

10.1.3 Trapezoidal profile

- $\alpha_{acc} = \omega_{max} / t_{acc}$; $\alpha_{dec} = -\omega_{max} / t_{dec}$; phases t_{acc}, t_{const}, t_{dec}, t_{stop}

10.1.4 Inertia

- Arm as rod about pivot: $J_{arm} = \frac{1}{3} \cdot m_{arm} \cdot L^2$
- Ball as point mass at tip: $J_{ball} = m_{ball} \cdot L^2$
- Off-centroid pivot (parallel-axis theorem): $I_{pivot} = I_{COM} + m \cdot d^2$
- Total: $J_{total} = J_{arm} + J_{ball}$ (+ J_{rotor}; if geared, $J_{eq} = J_{load}/n^2 + J_{rotor}$)

10.1.5 Phase torques

- Inertial: $T_\alpha = J_{total} \cdot \alpha$
- Gravity (arm horizontal, worst case): $T_{grav} = g \cdot (m_{arm} \cdot L/2 + m_{ball} \cdot L)$
- Acceleration phase: $T_{acc} = T_\alpha + T_{grav}$; Constant: $T_{const} = T_{grav}$; Deceleration: $T_{dec} = T_{grav} - T_\alpha$

10.1.6 RMS torque & safety factor

- $T_{rms} = \sqrt{[\ (\Sigma\ T_i^2 \cdot t_i) / t_{total}\ ]}$
- $T_{peak,final} = 1.5 \cdot T_{acc}$; $T_{rms,final} = 1.5 \cdot T_{rms}$

10.1.7 L298N DC-motor electrical sizing (shooter)

- Current: I = T_peak,final / K_t (K_t = torque constant)
- Terminal voltage: V = I·R_a + K_e·ω_max (R_a = winding resistance, K_e = back-EMF constant)
- PWM duty from 12 V rail: D = V / 12 → analogWrite = round(D·255)
- Power / efficiency: P_mech = T·ω ; P_elec = V·I ; η = P_mech / P_elec

10.1.8 Servo adequacy check (lift & sorting)

- Condition: T_peak,final < T_stall(servo) ; margin = T_stall / T_peak,final

10.2 Shooting Mechanism — Projectile

The difference between launch and landing height is 12.2 cm and an optimal 40° launch angle, the required launch velocity for a 38 cm range follows from the flat-ground range equation.

Quantity	Symbol	Value
Horizontal range	R	0.38 m
Launch angle	θ	40°
Gravity	g	9.81 m/s²
Launch velocity	v = √(R·g / sin 2θ)	2.48 m/s
Time of flight	t = 2v·sinθ / g	0.2 s
Maximum height	y_max = (v·sinθ)² / 2g	0.1295 m

W=1.61875 rad/s^2

The difference between launch and landing height is 12.2 cm and an optimal 40° launch angle, the required launch velocity for a 48 cm range follows from the flat-ground range equation.

Quantity	Symbol	Value
Horizontal range	R	0.48 m
Launch angle	θ	40°
Gravity	g	9.81 m/s²
Launch velocity	v = √(R·g / sin 2θ)	2.62 m/s
Time of flight	t = 2v·sinθ / g	0.239 s
Maximum height	y_max = (v·sinθ)² / 2g	0.1446 m

W=1.8075 rad/s^2

_By taking average for the angular speeds

W_avg= 1.713125 rad/s^2

10.3 Actuator Sizing — Shooting (DC Motor)

The launching arm releases the ball following a trapezoidal velocity profile. Using CAD-derived inertias:

Quantity	Value
Total system inertia J_sum	5.642×10^{-6} kg·m²
Angular acceleration α	3.42 rad/s²
Resisting (gravity) torque T_grav	4.15 mN·m
Peak torque T_peak	≈ 4.15 mN·m (0.00623 N·m with SF = 1.5)
RMS torque T_rms	≈ 4.15 mN·m (0.006225 N·m with SF = 1.5)

10.4 Actuator Sizing — Sorting Servos

10.4.1 Upper Sorting Servo (Servo 1)

Inertia computed via the parallel-axis theorem with a 15% structural factor, over a trapezoidal profile (ω_max = 4.189 rad/s).

Quantity	Raw value	With SF = 1.5
System inertia J_sum	9.5202×10^{-6} kg·m²	—
Peak torque T_peak	7.48×10^{-5} N·m	1.495×10^{-4} N·m
RMS torque T_RMS	3.52×10^{-5} N·m	7.05×10^{-5} N·m

10.4.2 Lower Sorting Servo (Servo 2)

Quantity	Raw value	With SF = 1.5
System inertia J_sum	1.069×10^{-5} kg·m²	—
Peak torque T_peak	1.120×10^{-4} N·m	1.680×10^{-4} N·m
RMS torque T_RMS	5.279×10^{-5} N·m	7.919×10^{-5} N·m

10.5 Actuator Sizing — Lifting Servo

The lift raises the ball through a vertical displacement on a trapezoidal profile (ω_max = 4.189 rad/s, 3.0 s cycle), combining load and motor inertia.

Quantity	Raw value	With SF = 1.5
Total system inertia J_total	2.7967×10^{-5} kg·m²	—
Peak torque T_peak	2.929×10^{-4} N·m	4.39×10^{-4} N·m
RMS torque T_RMS	1.3807×10^{-4} N·m	2.07×10^{-4} N·m

These torque demands are small relative to the rated stall torque of typical hobby servos, confirming the selected servos are adequately sized with margin. State the actual servo model(s) used and their rated torque to close the comparison.

10.6 Current Consumption & Power Sizing

Estimate the per-subsystem current draw and the total system current under worst case (all actuators active), then size the SMPS and buck converter accordingly. This directly addresses the “subsystem current consumption” and “power sizing” deliverables.

System 1 — Sorting

Component	Qty	Typ (mA)	Total (mA)	Peak (mA)
Arduino Uno (controller) active	1	50	**50**	60
IR Sensor — presence FC-51 type	1	20	**20**	24
IR Sensor — color FC-51 type	1	20	**20**	24
Servo (Upper) moving average	1	250	**250**	600
Servo (Lower) moving average	1	250	**250**	600
LCD 16×2 I2C backlight on	1	30	**30**	36
Push Buttons ×4 each	4	1	**4**	5
Subtotal			**624 mA**	1349 mA

System 2 — Shooting

Component	Qty	Typ (mA)	Total (mA)	Peak (mA)
IR Sensor — presence FC-51 type	1	20	**20**	24
L298N Motor Driver quiescent	1	36	**36**	43
DC Motor (via L298N) running load	1	600	**600**	2100
Limit Switch pull-up resistor	1	1	**1**	1
Subtotal			**657 mA**	2168 mA

System 3 — Lifting

Component	Qty	Typ (mA)	Total (mA)	Peak (mA)

R Sensor — presence FC-51 type	1	20	**20**	24
Servo (Lifter) moving average	1	250	**250**	600
Servo (Gate) moving average	1	250	**250**	600
Subtota			**520 mA**	1224 mA

Communication & Control

Component	Qty	Typ (mA)	Total (mA)	Peak (mA)
Arduino Mega ATmega2560 active	1	93	**93**	112
Arduino Uno (shared Sys1-3) active	1	50	**50**	60
HC-05 Bluetooth connected state	1	40	**40**	48
Subtotal			**183 mA**	220 mA

1984 mA	4961 mA	2480 mA
Typical total draw	Peak total draw	Recommended supply (×1.25)

Design Notes & Warnings

- Servo stall current can reach 1.5–2 A each. Do not rely on typical value during design.
- DC motor startup inrush can be 3–5x the running current. Size L298N heatsink accordingly.
- Buck converter efficiency ~85% — derate input supply: if load = 1.5 A @ 5 V, input needs ~900 mA @ 12 V.
- Arduino Uno is shared across Systems 1, 2, and 3 — counted once (50 mA) in Communication section.
- L298N has ~2 V internal drop — DC motor must be fed from 7–12 V rail, not 5 V.

Datasheet Reference

Component	Typical (mA)	Peak (mA)	Source
Arduino Uno	50	200	Atmel ATmega328P datasheet
Arduino Mega	93	200	ATmega2560 datasheet
SG90 / MG90 Servo	250	~600	Hobby servo datasheet
DC Motor (generic)	600	~2100	12 V DC motor typical
L298N Driver	36	70	ST L298N datasheet
HC-05 Bluetooth	40	50	HC-05 module spec
IR Sensor (FC-51)	20	30	FC-51 module spec
LCD 16×2 + I2C	30	80	Backlight enabled

11. Prototyping & Iteration

Document the prototyping journey: the materials and tools used, the versions of each subsystem that were built and tested, and the changes made in response to testing or feedback. The course rewards a visible, iterative design process, so capture the “before and after” of each significant change.

11.1 Iteration Log

Iteration	Change made	Reason / outcome
v1	first sorting prototype with slider mechanism in lifting	Slider mechanism didn’t work
v2	Vertical lifting	Easier/ did the job
v3	shooter speed& firing angle tuning	Reach the cone
Final	integrated build	

12. Safety Considerations

As a toy intended for children aged 2–10, safety governed many design decisions. The following measures address mechanical, electrical, and material safety; complete the validation column once inspections are done.

Area	Measure
Edges	No sharp edges; rounded/chamfered MDF/covered with 3d printed frame
Wiring	All wiring hidden inside the chassis
Moving parts	Servos/motor enclosed; pinch points covered
Electrical	Single low-voltage source; no exposed contacts
Stop control	Physical start/stop button accessible
Standards	Aligned with children's-toy safety (e.g. EN 71 / CE intent)

The system also has software safety behaviours: a Bluetooth-disconnection failsafe on the Mega, a remote reset from the app, and a reset routine that stops the motor and returns all servos to their home positions and clears the active sequence.

13. Educational Impact

Describe the developmental and educational benefits the toy delivers to its target age group. The reactor-defusal framing supports several learning areas:

- Colour recognition and discrimination (white vs black fuel cells).
- Sequencing and short-term memory (reproducing the generated code).
- Cause and effect and feedback (immediate physical and on-screen response to each choice).
- Pattern matching and basic logical reasoning under a time constraint.
- Exposure to automation concepts — sorting, transport, and dispatch — through a visible mechanical process.

Tie these benefits back to the findings from the user research in Section 3, and note any observations from letting children try the toy.

14. Project Management

14.1 Team Roles & Responsibilities

Member	Responsibility
Anas	CAD design
Omar	Coding + CAD design
Teefa	Application, communication, code integration of the 3 mechanisms
Essa	PCB, wiring diagrams, PDS, block diagram
Mario	Calculations + documentation

14.2 Work Breakdown Structure (WBS)

WBS	Task Name	Dur.(days)	Start	Finish
1.1	SOFTWARE & SPATIAL INTEGRATION		W1 7/3	W12 23/5
1.1.1	Sorting Subsystem		W1 7/3	W7 18/4
1.1.1.1	Idea & concept	3d	W1 7/3	W1 7/3
1.1.1.2	CAD design	7d	W1 7/3	W2 14/3
1.1.1.3	3D printing / fabrication	7d	W2 14/3	W3 21/3
1.1.1.4	Actuator sizing	5d	W3 21/3	W4 28/3
1.1.1.5	Arduino coding	7d	W4 28/3	W5 4/4
1.1.1.6	Wiring diagram	5d	W5 4/4	W6 11/4
1.1.1.7	Individual testing	3d	W6 11/4	W6 11/4
1.1.1.8	Subsystem integration	3d	W6 11/4	W7 18/4
1.1.2	Shooting Subsystem		W1 7/3	W7 18/4
1.1.2.1	Idea & concept	3d	W1 7/3	W1 7/3
1.1.2.2	CAD design	7d	W1 7/3	W2 14/3
1.1.2.3	3D printing / fabrication	7d	W2 14/3	W3 21/3
1.1.2.4	Actuator sizing + projectile calc	5d	W3 21/3	W4 28/3
1.1.2.5	Arduino coding	7d	W4 28/3	W5 4/4
1.1.2.6	Wiring diagram	5d	W5 4/4	W6 11/4
1.1.2.7	Individual testing	3d	W6 11/4	W6 11/4
1.1.2.8	Subsystem integration	3d	W6 11/4	W7 18/4
1.1.3	Lifting Subsystem		W1 7/3	W7 18/4
1.1.3.1	Idea & concept	3d	W1 7/3	W1 7/3
1.1.3.2	CAD design	7d	W1 7/3	W2 14/3
1.1.3.3	3D printing / fabrication	7d	W2 14/3	W3 21/3
1.1.3.4	Actuator sizing	5d	W3 21/3	W4 28/3
1.1.3.5	Arduino coding	7d	W4 28/3	W5 4/4
1.1.3.6	Wiring diagram	5d	W5 4/4	W6 11/4
1.1.3.7	Individual testing	3d	W6 11/4	W6 11/4
1.1.3.8	Subsystem integration	3d	W6 11/4	W7 18/4
1.1.4	Communication System		W5 4/4	W12 23/5
1.1.4.1	Mega ↔ Uno (I2C/UART) setup	5d	W5 4/4	W6 11/4
1.1.4.2	LCD + Buttons firmware (Mega)	7d	W4 28/3	W6 11/4
1.1.4.3	Bluetooth HC-05 setup	5d	W5 4/4	W7 18/4
1.1.4.4	Processing.io app (Display 2)	7d	W6 11/4	W8 25/4
1.1.4.5	Custom PCB design (Fritzing)	10d	W5 4/4	W8 25/4
1.1.4.6	Communication testing	3d	W8 25/4	W9 2/5
1.2	Design Documentation		W1 7/3	W2 14/3
1.2.1	PDS document	5d	W1 7/3	W2 14/3
1.2.2	WBS & GANTT	2d	W1 7/3	W1 7/3
1.2.3	System flowchart & block diagram	5d	W1 7/3	W2 14/3
1.3	Full Integration		W8 25/4	W12 23/5
1.3.1	Connect all subsystems on PCB	5d	W8 25/4	W9 2/5
1.3.2	Full marble closed-loop test	5d	W9 2/5	W10 9/5

Figure 14.1 — Work breakdown structure.

14.3 Gantt Chart (12-week)

Critical path: CAD → Firmware → PCB → Integration. App / Bluetooth runs in parallel and converges at integration (W7–8). ■ = active, ▫ = light / ongoing.

Workstream	Owner	W1–2	W3–4	W5–6	W7–8	W9–10	W11–12
Documentation + calc	Mario	■	▫	■	▫	▫	■
PDS / block / wiring	Essa	■	■				
Mechanical / CAD	Anas, Omar	■	■	■			
Firmware / coding	Omar, Teefa		■	■	■		
App / Bluetooth	Teefa			■	■	■	
PCB / Electrical	Essa		■	■	■		
Integration & test	All				■	■	■

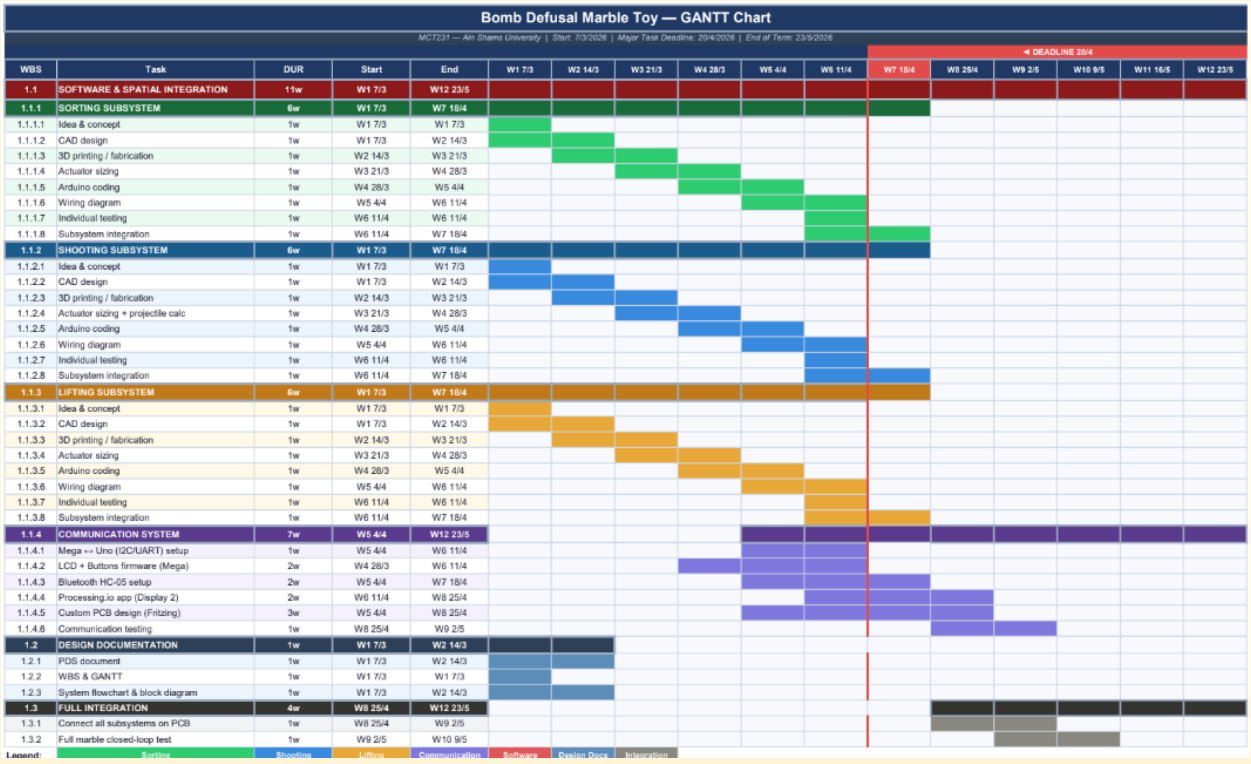


Figure 14.2 — Project schedule (Gantt).

14.4 Product Design Specification (PDS)

The PDS captures the binding requirements and how each will be validated.

#	Requirement area	Specification (summary)
1	Performance	Sorting accuracy ≥ 95%; actuators respond < 200 ms; near-zero polling latency
2	Safety	No accessible sharp edges (3D-printed edge covers on exposed MDF edges); hidden wiring; guarded moving parts; CE/EN 71 intent
3	Functional	Sorting + lifting + shooting operational; single entry point; start/stop
4	User interfaces	On-board LCD (timer/score) + Processing app via Bluetooth (game state)
5	MCUs & comms	≥ 2 Arduinos (Uno + Mega) over UART, 9600 baud; HC-05 Bluetooth to external app
6	Custom PCB	Two custom shield-PCBs (Uno + Mega) distributing rails with modular headers
7	Life span	≥ 3 years regular use; standard replaceable components
8	Maintenance	Modular hardware/code; each subsystem independently testable
9	Size & weight	Footprint within course limit; modular plug connections
10	Aesthetic	Reactor/space motif; red countdown; neat wiring; intuitive controls
11	Power	Single SMPS (220 V AC → 12 V DC); 12 V → L298N, buck → servos, 9 V reg → Uno; common GND on PCB
12	Manufacturing	Laser-cut MDF + FDM prints; CAD + electrical files in Git

Note: items 5, 6, and 11 reflect the final architecture — UART (not I2C) for the Mega–Uno link, two shield-PCBs, and a single SMPS source. Ensure the block diagram matches.

15. Bill of Materials (BOM)

Complete the BOM below with quantities, suppliers, and costs. This supports the deliverables list and the costing discussion.

#	Item	Spec / Model	Qty	Subsystem
1	Microcontroller (master)	Arduino Mega 2560	1	Control
2	Microcontroller (slave)	Arduino Uno R3	1	Control
3	Bluetooth module	HC-05	1	Comms
4	Display	I2C 16×2 LCD (0x27)	1	UI
5	Servo motors	[MG90S / SG90]	3	Sort/Lift
6	DC motor	-	1	Shooter
7	Motor driver	L298N H-bridge	1	Shooter
8	IR sensors	IR obstacle/colour	4	Sensing
9	Limit switch	SPDT micro	1	Shooter
10	Push buttons	White/Black/Reset/Submit	4	UI
11	Power supply	SMPS, 220 V AC → 12 V DC	1	Power
12	Buck converter	step-down → servo rail	1	Power
13	Voltage regulator	9 V linear (Uno VIN)	1	Power
14	Custom PCB — Uno shield	pin-header shield	1	Electrical
15	Custom PCB — Mega shield	pin-header shield	1	Electrical
16	Chassis	MDF, laser-cut	—	Mechanical
17	Tracks / funnels	FDM 3D print (navy)	—	Mechanical
18	3D-printed edge covers	FDM, edge guards	—	Mech/Safety
19	Wiring / connectors	sleeving, JST headers	—	Electrical

16. Assurance of Properties

Each property is assured in two steps. Validation confirms the design meets the user/goal intent; verification confirms it meets the measurable technical specification. Pass criteria align with the PDS targets in Section 14.4.

Category	Validation (goal)	Verification (spec / test)
Sorting	Correct colour cells advance the sequence; wrong cells are rejected	IR colour read matches ball ≥ 95% over 50 cycles; path servo hits 142° / 97° set-points
Shooting	Accepted ball launched cleanly into the funnel	DC motor fires at PWM 90/255; lands in funnel ≥ 95%; limit-switch re-home each shot
Lifting	Every ball returns to the start with no manual reload	Lift IR detects ball; servo 0°→95°→0°; 0 dropped balls over 50 cycles
Closed-loop integration	Continuous unattended sort→shoot/return→lift cycle	50+ uninterrupted cycles, no jam/double-feed; slot advances only on match
Mega–Uno comms	Boards stay in step through the whole game	UART 9600 baud; 0 dropped WBW… / R / D messages on serial monitor
HC-05 + App sync	App code, countdown, and lives mirror the physical machine	Pairing < 5 s; W/B/R→Mega and C/W→App verified; latency < 200 ms; BT-drop triggers reset
LCD interface	Player can read target length, input, and result	Real-time update of input slots and SUCCESS/FAIL states
Power integrity	Single SMPS source runs all subsystems without fault	SMPS 220 V→12 V; 12 V→L298N, buck→servos, 9 V reg→Uno; rail droop < 5%, no overheat (30 min)
Safety — sharp edges	No sharp MDF edges reachable by a child	3D-printed edge covers fitted to exposed laser-cut MDF edges (edge guards only, not a full enclosure); inspected for full coverage of accessible edges
Safety — wiring/PCB	No exposed wires or accessible electrical contacts	Harness routed below deck; shield-PCBs enclosed; continuity / short check passed
Safety — moving parts	Safe for ages 2–10	Moving parts guarded; choking-hazard review on balls
Modularity	Subsystems swap quickly for repair	Each module on keyed plate + single connector; swap < 5 min, no tools

17. Final Product

Present the completed toy. Include high-quality images from several angles, and reference the demonstration video.



Figure 17.1 — Final assembled product.

Demonstration video:

